Assignment 2

1) **MD simulation of a protein with different force fields**

   As we learned through out the course, an empirical potential energy function, called force field, is used to describe the interactions between atoms in molecular dynamics (MD) simulations. In the MD practical of the small B1 domain of protein G, we chose one of the conventionally used force fields but we could have chosen a different one. In this assignment we will compare the force field we used originally for the practical (OPLS-AA/L all-atom for the protein and the SPC water model) with other force fields, one grouping few atoms into particles, named GROMOS, and another derived using machine learning, called GRAPPA.

Assignment data science and simulations course (SS 2026)

Matrikel-Nr 1: **Matrikel-Nr 2:**

- Please indicate if you need a grade (1–6) or a passed/failed (bestanden/nicht bestanden) confirmation
- Please submit only one single PDF file per team (not one per team member) containing the answers to the questions and the relevant data and plots to support them (minimum 2 pages and maximum 6 pages).
- Please submit the assignment on July 26, 2026, by email to apontec@mpip-mainz.mpg.de

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Assignment 2: MD simulation of a protein with different force fields

Introduction

As we learned through out the course, an empirical potential energy function, called force field, is used to describe the interactions between atoms in molecular dynamics (MD) simulations. In the MD practical of the small B1 domain of protein G, we chose one of the conventionally used force fields but we could have chosen a different one. In this assignment we will compare the force field we used originally for the practical (OPLS-AA/L all-atom for the protein and the SPC water model) with other force fields, one grouping few atoms into particles, named GROMOS, and another derived using machine learning, called GRAPPA.

General indications

1) Consider the MD practical of a small protein:
   `https://colab.research.google.com/drive/19Vb28lg-o_mZ9Ylpzh7uAJbmaTqWOWKm?usp=sharing`
2) To observe the effect more clearly, increase the length of the simulation from 20 ps to 2 ns, by changing `nsteps` in the file `md.mdp` (keep in mind that the integration time step is `dt = 0.002 ps`).
3) For comparison, collect all the results with more steps—with the original used force field (OPLS-AA/L all-atom for the protein and the SPC water model)—.
4) Repeat the setup , but this time, when running `gmx pdb2gmx` choose

   `GROMOS96 43a1` for the protein and

   `SPC` for the water model
5) Run the 2-ns MD simulation with this different force field.

   How long these simulations run will depend on the computer. In a standard 4-core laptop they should take around 1.5 hour.
6) Repeat this setup using the machine-learned molecular-mechanics force field GRAPPA. To this end follow these instructions:
   a) Using `gmx pdb2gmx`, create an initial topology choosing the AMBER99SB-ILDN protein force field and TIP3P water model.
   b) Convert the topology to the GRAPPA topology. This would be done by installing GRAPPA and using the command `grappa_gmx`. However, we have converted the topology for you so you do not need to install this package. Please download the output file, directly in your Google Colab notebook by running:

   ```
   wget https://mptg-cbp.github.io/teaching/tutorials/protein/topology_grappa.top
   ```

c) Continue the workflow as for the other force fields, but using the GRAPPA topology file `topology_grappa.top` instead of `topol.top`, for **every** `gmx` command that takes a `.top` file as input.

Recommendation: carry out the tutorial in separate directories so files are not overwritten.

Questions (please explain your answers and include the necessary data and plots to support them):

1) General outcome of the simulation:
   - How many integration steps were needed to run the 2-nanosecond simulation?
   - GROMOS is a so-called a united-atom force field. Some moeities (e.g. CH3 groups) are treated as single particles. How many particles less does this system have compared to the OPLS-AA one?
   - What is the performance gain by using the united-atom approach (tip: check the Performance line in the md.log files)?
2) How stable is the protein in the three used force fields, in terms of RMSD and radius of gyration?
3) Determine if the use of the united-atom GROMOS force field or the machine-learning GRAPPA force field caused a change in the flexibility of certain parts of the protein, monitoring the RMSF, and if that is the case, explain why. How could you further evaluate your hypothesis?
4) How does the potential energy compare in the three cases? Provide an explanation of the observed behavior.
5) What can you say about the hydrophobic area exposed to the solvent for the three used force fields?
6) Why do you think it is important to check different force fields in MD simulations?
7) How can machine learning improve force fields for MD simulations?